=== UNIT 6: TECHNICAL VOCABULARY, PHRASAL VERBS & PROFESSIONAL STYLE ===

>> 6.1 Academic Phrasal Verbs in Scientific Discourse

A phrasal verb consists of a base verb combined with a preposition or adverbial particle, creating an idiomatic meaning distinct from its component words. In formal academic writing, single-word formal Latinate verbs are frequently preferred over informal phrasal verbs.

* Common Academic Phrasal Verbs and Formal Latinate Equivalents:

- Break down: Decompose, disintegrate, or fail mechanically. (Formal: Decompose / Degrade).
- Carry out: Execute or perform an experiment. (Formal: Conduct / Execute / Perform).
- Figure out: Solve or comprehend a complex problem. (Formal: Deduce / Determine / Resolve).
- Look into: Investigate or examine thoroughly. (Formal: Investigate / Examine).
- Put off: Postpone or delay an activity. (Formal: Defer / Postpone).
- Set up: Assemble apparatus or organize an assay. (Formal: Assemble / Establish / Configure).
- Point out: Draw attention to a specific detail. (Formal: Indicate / Highlight).
- Bring about: Cause something to happen. (Formal: Induce / Cause / Precipitate).
- Phase out: Gradually discontinue use. (Formal: Eliminate / Discontinue).

>> 6.2 Commonly Confused Words (Homophones and Paronyms)

* Affect vs. Effect:

- Affect (Verb): To influence or alter: "Substrate concentration affects enzymatic velocity."
- Effect (Noun): The outcome or result: "The Pasteur effect describes glycolytic inhibition by oxygen." (Exception: Effect as verb means ’to bring about’: effect change).

* Complement vs. Compliment:

- Complement: That which completes, enhances, or pairs harmoniously: "The two primers complement the target DNA sequence."
- Compliment: An expression of praise or admiration: "The reviewer paid a generous compliment."

* Principal vs. Principle:

- Principal: Primary, foremost, or chief executive of an institution: "The principal investigator secured the grant."
- Principle: A fundamental scientific law, doctrine, or moral rule: "Le Chatelier’s principle predicts equilibrium shifts."

* Discrete vs. Discreet:

- Discrete: Distinct, separate, discontinuous: "The data points fall into four

discrete clusters."
- Discreet: Prudent, unobtrusive, tactful: "The technician remained discreet regarding patent disclosures."
* Ensure vs. Insure vs. Assure:
- Ensure: To guarantee or make certain: "Ensure the autoclave reaches 121 deg C."
- Insure: To protect against financial loss via policy.
- Assure: To remove doubt from a person through verbal confirmation: "I assure you the calculations are accurate."

>> 6.3 Professional Idioms and Metaphorical Registers

While colloquial idioms should be minimized in research manuscripts, high-level metaphorical idioms appear frequently in professional engineering management:
* Back to the drawing board: Restart a failed project from the initial design phase.
* Cut corners: Undertake procedures unsafely or cheaply at the expense of quality.
* Burn the midnight oil: Work late into the night on demanding technical projects.
* Hit the nail on the head: Identify the precise core cause of an engineering anomaly.
* At the cutting edge / state of the art: Operating at the foremost frontier of technology.

[EXERCISE 1: Formal Latinate Replacement of Informal Phrasal Verbs]
Problem: Replace the informal phrasal verbs in the following sentences with precise, single-word formal academic equivalents:
Sentence 1: "The research team carried out [_____] the genetic sequencing protocol."
- Formal Replacement: CONDUCTED / EXECUTED / PERFORMED.
Sentence 2: "The organic contaminants broke down [_____] within forty-eight hours."
- Formal Replacement: DECOMPOSED / DEGRADED.
Sentence 3: "The engineering committee will look into [_____] the pump failure."
- Formal Replacement: INVESTIGATE / EXAMINE.
Sentence 4: "The principal investigator pointed out [_____] the discrepancy in the absorbance values."
- Formal Replacement: INDICATED / HIGHLIGHTED.
Sentence 5: "We must figure out [_____] the mechanism governing membrane transport."
- Formal Replacement: ELUCIDATE / DETERMINE / DEDUCE.

[EXERCISE 2: Homophone and Confused Word Disambiguation Drill]
Problem: Select the correct lexical item in each of the following engineering sentences:
Sentence 1: "The addition of heavy metal ions adversely (affected / effected) cell viability."
- Solution: AFFECTED. (Verb meaning 'influenced').
Sentence 2: "The spectrophotometer works on the fundamental (principal / principle) of

Beer-Lambert’s Law."
- Solution: PRINCIPLE. (Noun meaning ’fundamental scientific law’).
Sentence 3: "Electrophoresis separated the digested DNA into five (discrete / discreet) bands on the agarose gel."
- Solution: DISCRETE. (Adjective meaning ’distinct and separate’).
Sentence 4: "The bio-sensor electrode features a self-assembled monolayer that (complements / compliments) the enzyme cofactor."
- Solution: COMPLEMENTS. (Verb meaning ’pairs with and completes’).
Sentence 5: "Sterilization indicators are placed inside the chamber to (ensure / insure / assure) complete thermal penetration."
- Solution: ENSURE. (Verb meaning ’to make certain’).

[EXERCISE 3: Precision Academic Sentence Rewriting]
Problem: Eliminate wordiness, colloquialisms, and redundant phrasing in the following excerpt:
Wordy Draft: "At this point in time, it is really totally obvious that the reason why the reaction failed was because of the fact that the technician didn’t bother to check the temperature."
Concise Academic Revision: "The reaction failed because the technician did not verify the operating temperature."
Linguistic Analysis:
- ’At this point in time’ -> Eliminated (redundant fluff).
- ’really totally obvious’ -> Eliminated (colloquial intensifiers).
- ’the reason why ... was because of the fact that’ -> Reduced to a single clean conjunction: ’because’.
- ’didn’t bother to check’ -> Upgraded to professional diction: ’did not verify’.

[EXERCISE 4: Contextual Vocabulary Synthesis in Biotechnology Context]
Problem: Match each technical definition with its correct academic term (Elucidate, Inhibit, Titrate, Lyophilize, Precipitate):
1. To determine the concentration of a dissolved substance by adding measured volumes of a standardized reagent: TITRATE.
2. To freeze-dry a biological sample under high vacuum to preserve protein stability: LYOPHILIZE.
3. To make lucid, clear, or explain a complex biochemical pathway: ELUCIDATE.
4. To hinder, restrain, or suppress an enzymatic reaction: INHIBIT.
5. To separate an insoluble solid substance from a solution: PRECIPITATE.

>> 6.4 University Review & Examination Questions

* Q1: Contrast the definitions, grammatical functions, and illustrative sentences of the five most commonly confused word pairs in technical writing.
Answer: (1) Affect (verb: to alter/influence: ’UV radiation affects DNA replication’) vs. Effect (noun: the consequence: ’The greenhouse effect raises atmospheric temperatures’). (2) Principal (adjective/noun: primary/chief executive: ’The principal

investigator authorized the purchase') vs. Principle (noun: fundamental natural law: 'The Heisenberg uncertainty principle limits precision'). (3) Discrete (adjective: separate/distinct: 'Quantum systems occupy discrete energy levels') vs. Discreet (adjective: tactful/prudent: 'The auditor maintained discreet confidentiality'). (4) Complement (verb/noun: that which completes: 'Guanine complements cytosine') vs. Compliment (verb/noun: expression of praise: 'The dean complimented the graduating cohort'). (5) Ensure (verb: to guarantee certainty: 'Ensure valves are closed') vs. Insure (verb: financial underwriting: 'The facility is insured against fire').

* Q2: Explain the stylistic differences between formal academic English and informal conversational English regarding phrasal verbs, contractions, and passive structures. Answer: (1) Phrasal Verbs vs. Latinate Vocabulary: Conversational English relies heavily on multi-word phrasal verbs (carry out, set up, figure out). Formal academic English replaces them with single-word precise Latinate verbs (conduct, assemble, elucidate) to enhance lexical density and avoid ambiguous idiomatic interpretations. (2) Contractions: Conversational English routinely contracts auxiliary verbs and negations (don't, can't, it's). Formal academic writing strictly forbids contractions, requiring full spelled-out forms ('do not', 'cannot', 'it is') to preserve structural formality. (3) Personal Pronouns vs. Passive Voice: Conversational English relies on subjective pronouns ('We did this', 'I observed that'). Formal scientific English uses passive constructions and impersonal nominalizations ('The assay was performed', 'Analysis revealed') to maintain objective, reader-independent neutrality.

>> 6.5 Key Takeaways and Revision Points

* Replace informal phrasal verbs with formal Latinate verbs (carry out -> conduct; look into -> investigate).
* Affect is a verb; Effect is a noun; Principal is chief/primary; Principle is a fundamental law.
* Discrete means separate; Discreet means prudent; Complement means that which completes.
* Professional technical style demands conciseness, precision, and elimination of colloquial wordiness.